Practical no. 5B

Aim: Create the following application:
b. Create a Product Table in SQL Server and perform Insert Operation using Entity Framework Core.

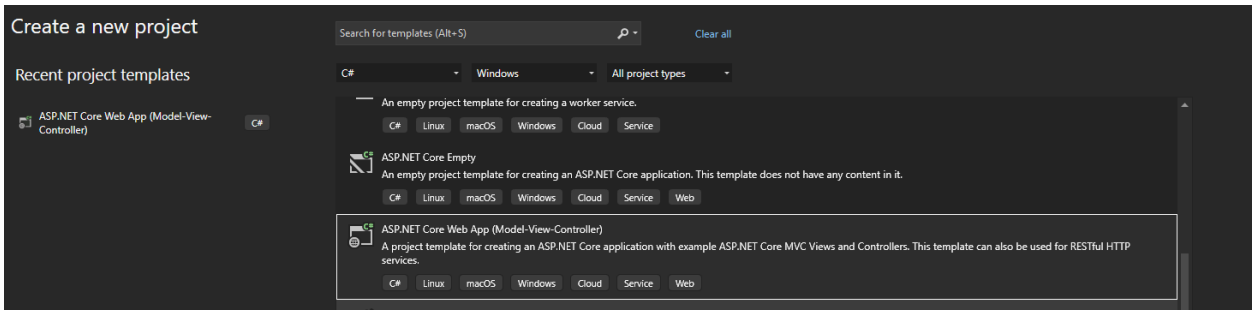
Software Required:

Visual Studio 2026

Procedure:

Step 1: Create New MVC Project

1. Open Visual Studio 2022.
2. Click Create a New Project.
3. Select ASP.NET Core Web App (Model-View-Controller).



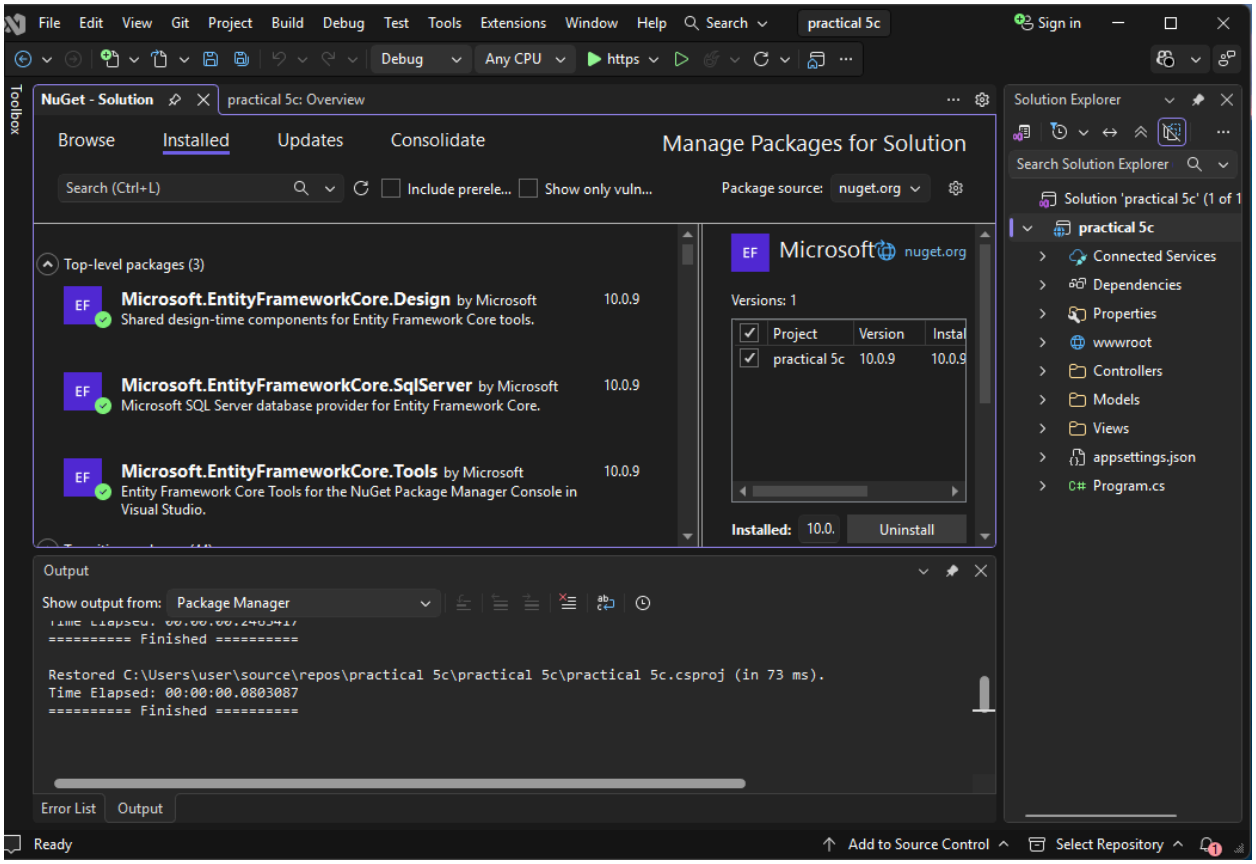
4. Click Next.
5. Enter Project Name: Practical5b
6. Select .NET 8.0.
7. Click Create.

Step 2: Install packages

Tools → NuGet Package Manager → Manage NuGet Packages

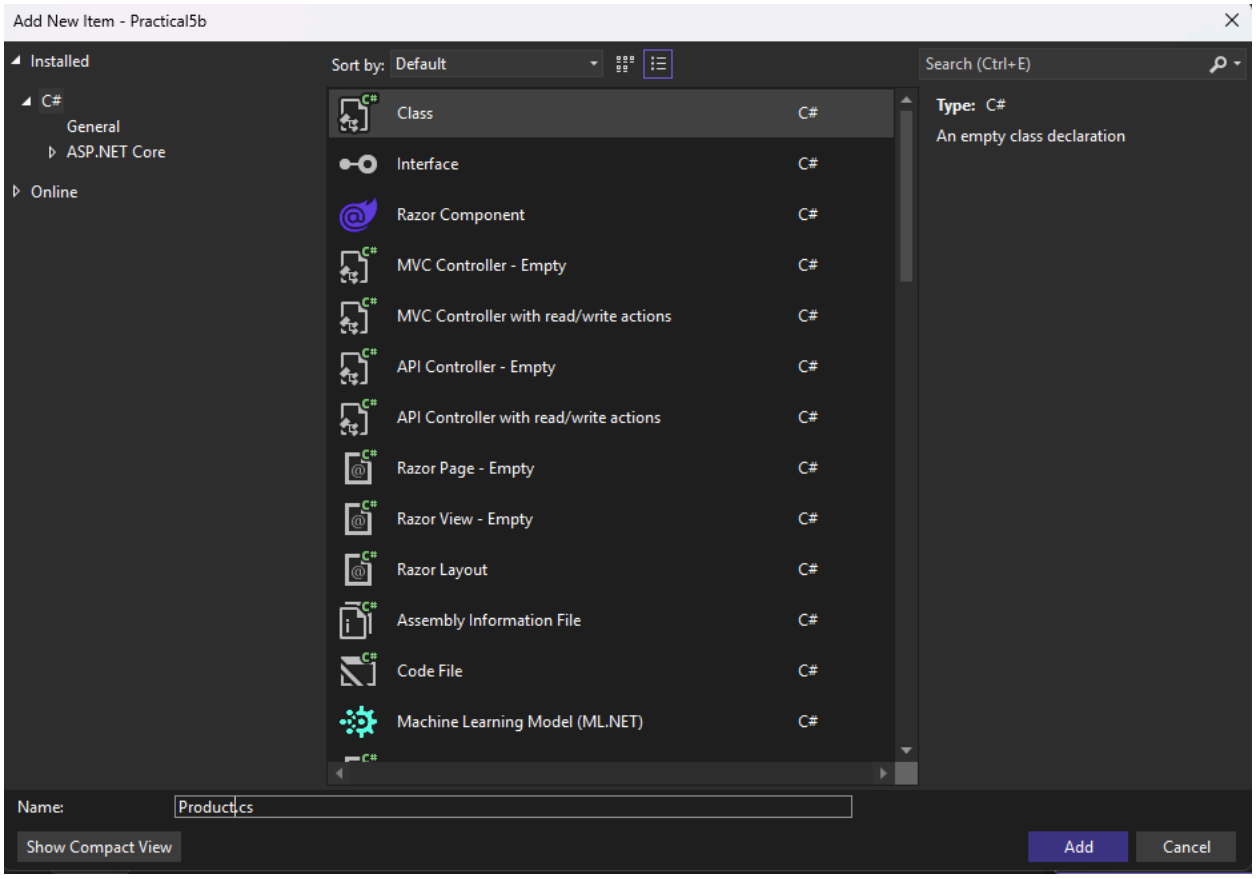
Install:

1. Microsoft.EntityFrameworkCore.Design
2. Microsoft.EntityFrameworkCore.SqlServer
3. Microsoft.EntityFrameworkCore.Tools



Step 3: Create Product Model

1. Right-click Models folder.
2. Add → New Item → Class.
3. Name it: Product.cs
4. Click Add.



Product.cs

```
using System.ComponentModel.DataAnnotations;

namespace Practical5b.Models
{
   public class Product
   {
      public int Id { get; set; }

      [Required]
      public string ProductName { get; set; }

      public decimal Price { get; set; }

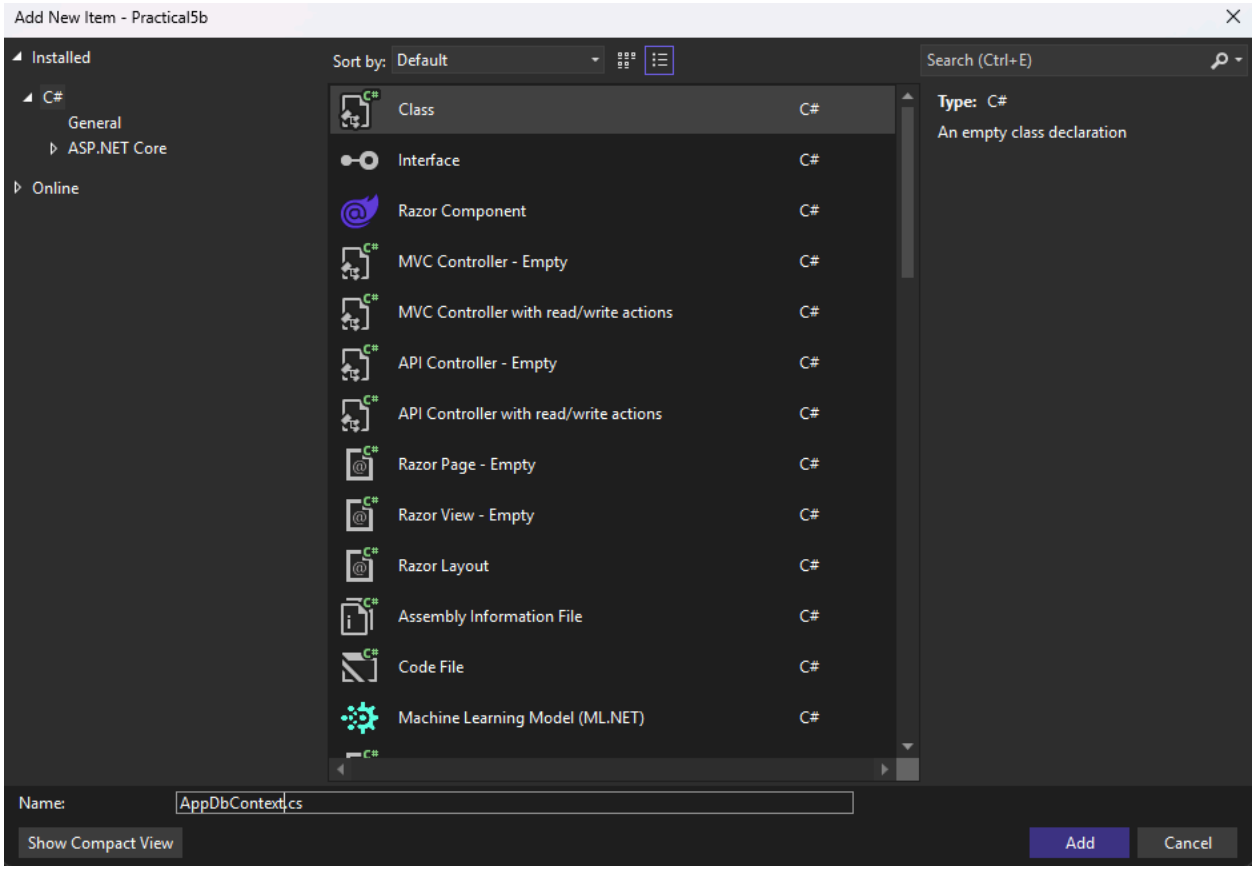
      public int Quantity { get; set; }
   }
}
```

Step 4: Create Data Folder

1. Right-click project.
2. Add → New Folder.
3. Name it: Data

Step 5: Create DbContext Class

1. Right-click the Data folder.
2. Add → New Item → Class.
3. Name it: AppDbContext.cs



AppDbContext.cs

using Microsoft.EntityFrameworkCore;
using Practical5b.Models;

namespace Practical5b.Data
{

```
    public class AppDbContext : DbContext
    {
        public AppDbContext(DbContextOptions<AppDbContext> options)
            : base(options)
        {
        }

        public DbSet<Product> Products { get; set; }
    }
}
```

Step 6: Configure Database Connection String
Open Solution Explorer → appsettings.json.

appsettings.json

```
{
  "ConnectionStrings": {
    "ProductConnection":
"Server=(localdb)\\MSSQLLocalDB;Database=ProductDB;Trusted_Connection=True;Tru
stServerCertificate=True;"
  },

  "Logging": {
    "LogLevel": {
      "Default": "Information",
      "Microsoft.AspNetCore": "Warning"
    }
  },

  "AllowedHosts": "*"
}
```

Step 7: Configure EF Core in Program.cs
Open Solution Explorer → program.cs.
program.cs

```
using Microsoft.EntityFrameworkCore;
using Practical5b.Data;
```

```
var builder = WebApplication.CreateBuilder(args);

// Add MVC Services
builder.Services.AddControllersWithViews();

// Add Database Context
builder.Services.AddDbContext<AppDbContext>(options =>
    options.UseSqlServer(
        builder.Configuration.GetConnectionString("ProductConnection")));

var app = builder.Build();

// Configure HTTP Pipeline
if (!app.Environment.IsDevelopment())
{
    app.UseExceptionHandler("/Home/Error");
    app.UseHsts();
}

app.UseHttpsRedirection();

app.UseStaticFiles();

app.UseRouting();

app.UseAuthorization();

app.MapControllerRoute(
    name: "default",
    pattern: "{controller=Product}/{action=Create}/{id?}").WithStaticAssets();

app.Run();
```

Step 8: Create Database Using Migration

Tools → NuGet Package Manager → Package Manager Console

Run following commands in power shell:

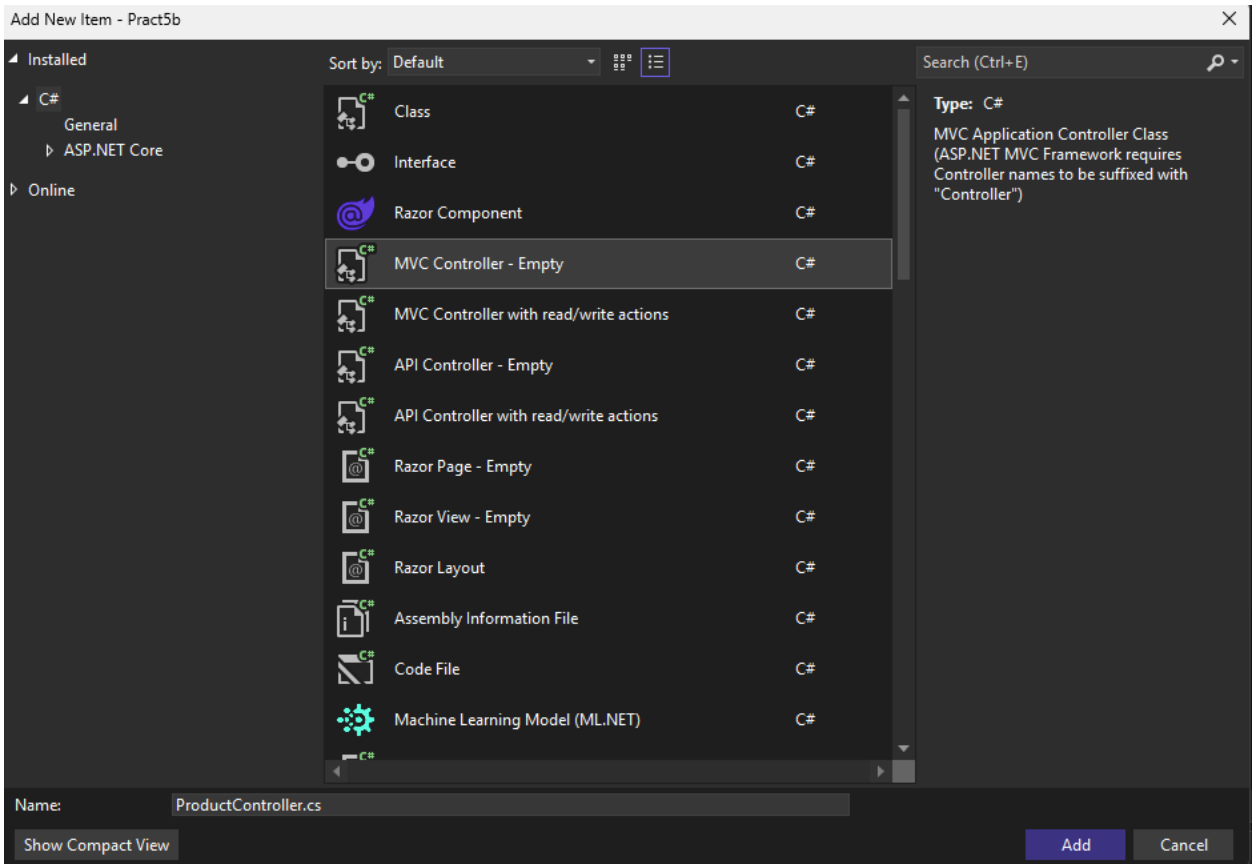
1. Add-Migration InitialCreate
2. Update-Database

EF Core automatically creates: ProductDB Database and Products Table.

Step 9: Create ProductController

Create a ProductController inside the Controllers folder.

1. Right Click Controllers.
2. Click Add.
3. Click New item→ MVC Controller -Empty
4. Name it: ProductController.cs
5. Click Add.



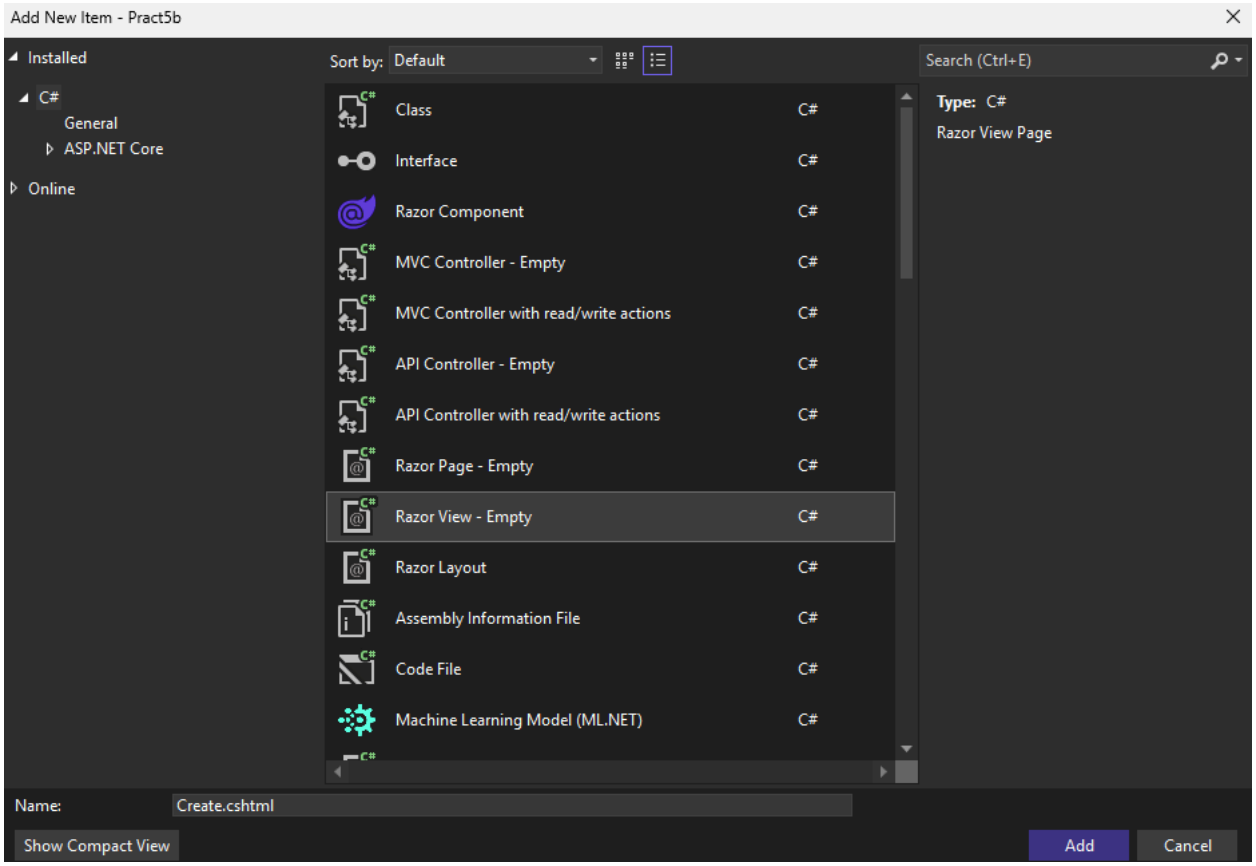

```
using Microsoft.AspNetCore.Mvc;
using Pract5b.Models;
using Pract5b.Data;
namespace Pract5b.Controllers
{
   public class ProductController : Controller
   {
```

```
        private readonly AppDbContext _context;
        public ProductController(AppDbContext context)
        {
            _context = context;
        }
        [HttpGet]
        public IActionResult Create()
        {
            return View();
        }

        [HttpPost]
        public IActionResult Create(Product product)
        {
            if (ModelState.IsValid)
            {
                _context.Products.Add(product);
                _context.SaveChanges();
                ViewBag.Message = "Product inserted successfully!";
                ModelState.Clear();
            }
            return View();
        }
    }
}
```

Step 9: Create View

1. Right-click View Folder.
2. Add → New Folder.
3. Name it: Product
4. Right-click the Product folder.
5. Add → New Item → razor View-Empty
6. Name it: Create.cshtml




```
@model Pract5b.Models.Product

@{
    ViewData["Title"] = "Create";
}

<h2>Add Product</h2>

<form asp-action="Create" method="post">

    <div>
        <label>Product Name</label>
        <input asp-for="ProductName" />
    </div>

    <br />

    <div>
        <label>Price</label>
```

```
        <input asp-for="Price" type="number" step="0.01" />
    </div>

    <br />

    <div>
        <label>Quantity</label>
        <input asp-for="Quantity" type="number" />
    </div>

    <br />

    <button type="submit">Insert Product</button>

</form>

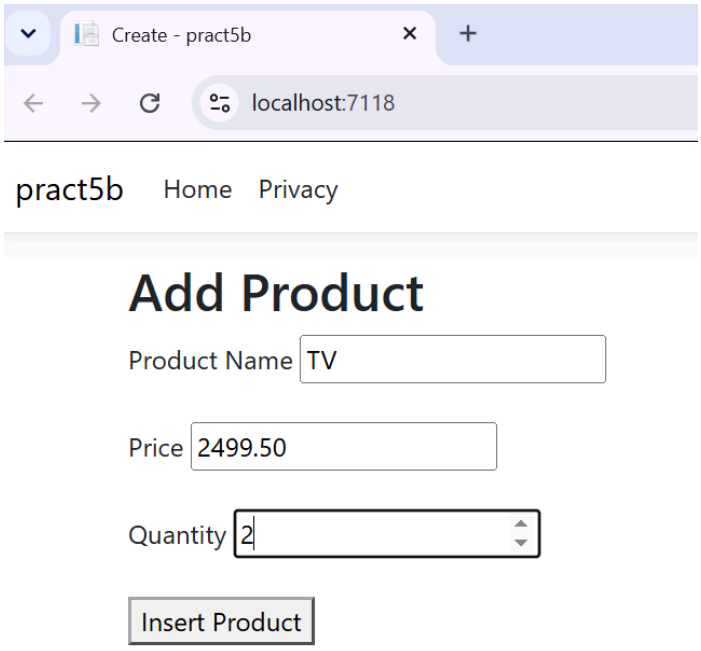
@if (ViewBag.Message != null)
{
    <p>@ViewBag.Message</p>
}
```

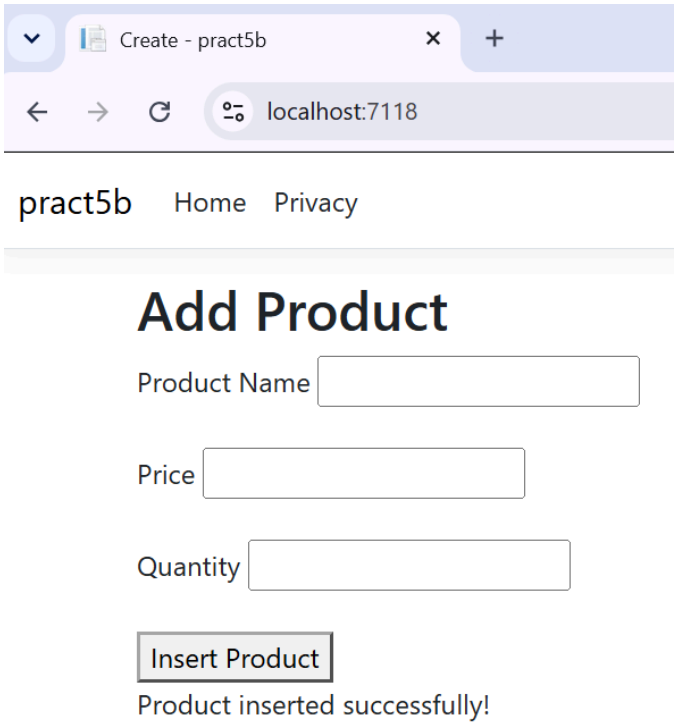
Step 10: Build Project

Ctrl + Shift + B

Step 11: Run Application

Press F5.


Create - pract5b
localhost:7118
pract5b
Home
Privacy
Add Product
Product Name
TV
Price
2499.50
Quantity
2
Insert Product


Create - pract5b
localhost:7118
pract5b
Home
Privacy
Add Product
Product Name
Price
Quantity
Insert Product
Product inserted successfully!